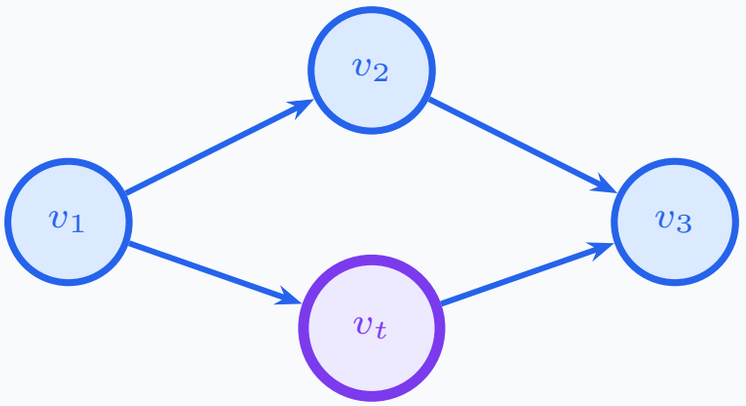
(1) GATv2 message passing



target endpoint $= v_t$

k-hop subgraph $\mathcal{G}_t$

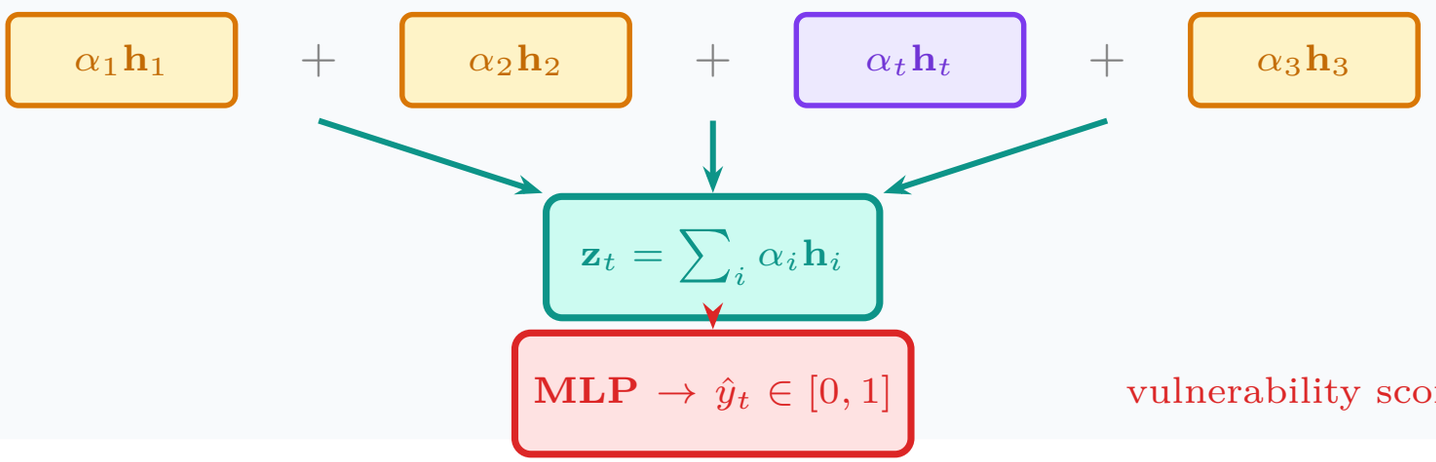
Multi-head attention on node + edge features

(2) Target-aware attention

$$\alpha_i = \frac{\exp\left(\mathbf{a}^\top [\mathbf{h}_i \| \mathbf{h}_t]\right)}{\sum_{j \in \mathcal{G}_t} \exp\left(\mathbf{a}^\top [\mathbf{h}_j \| \mathbf{h}_t]\right)}$$

Nodes structurally relevant to v_t's security context receive **higher** α_i

(3) Weighted readout



vulnerability score per endpoint